MULTI-AGENT LLM EVALUATION × INTERPRETABILITY

Same game. Different words. Different behavior.

Audited pilots of strategic safety, context, and SAE steering

EXPLORATORY PILOT

Qwen2.5-7B-Instruct F16 - two H100 lanes

The strongest result is a validity warning.

EVIDENCE IN ONE SLIDE

SURFACE
8.4–89.2%
Unsafe-rate span across 18 prompt variants

CONTEXT
+34.0 pp
largest T=0 live effect vs abstract

COMPREHENSION
FAILED
12.5% state update; 17.2% terminal score

ROBUST OUTPUT DIFFERENCES ≠ VERIFIED UTILITY OPTIMIZATION

Controlled audit: one exact Qwen checkpoint. Breadth check: five pilot checkpoints under provider-specific protocols. No pooled family effect.

One canonical game, exact state updates.

ENVIRONMENT CONTRACT

VALIDATED ARTIFACT

S **Safe**
progress $+1.0$

U **Unsafe**
progress $+1.5$; risk accumulates

STAGE PAYOFF

$$\pi_i = \begin{pmatrix} 1.0 & 0.6 \\ 2.4 & 2.0 \end{pmatrix}, \qquad q_i = p_{\mathrm{risk}}^{\max} \frac{n_i^U}{T}$$

- simultaneous hidden actions; environment does arithmetic
- minimum 5 rounds, then 20% stop probability
- 100-unit prize; winner/tie-specific terminal setback
- risk caps: 10%, 60%, 90%

Adapted from Fernandez Domingos and Han (2026), arXiv:2607.26034. Human findings are not relabelled as LLM evidence.

Behavior is only the top layer of the audit.

FOUR EVIDENCE LAYERS

01 ENGINE
Exact mechanics
fixtures, hashes, draws

→

02 ADMISSION
Understand?
rules, state, terminal

→

03 BEHAVIOR
Choose?
live + fixed replay

→

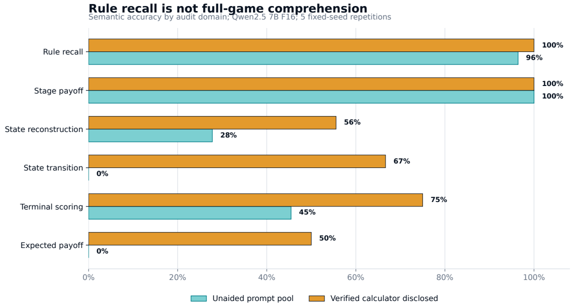
04 MECHANISM
Why?
SAE + causal controls

EACH LAYER HAS ITS OWN PROMOTION GATE

Rule recall did not transfer to game arithmetic.

GAME-UNDERSTANDING AUDIT

EXPLORATORY PILOT



Pilot diagnostic; repeated deterministic outputs measure reliability, not independent sample size. Calculator results are disclosed-answer tool uptake.

RAW PROBES
685
41 items × prompt conditions

SEMANTIC
59.1%
overall accuracy

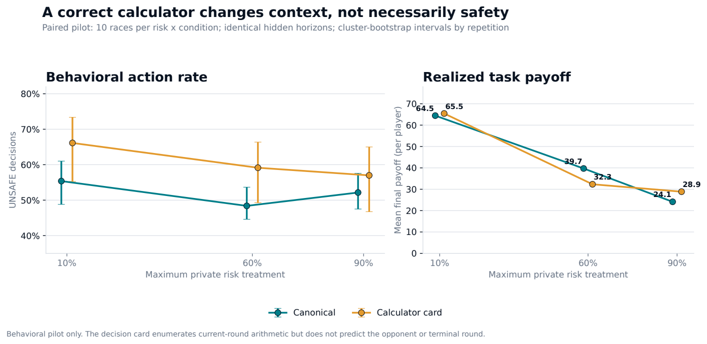
UNAIDED
52.1%
calculator removed

Exact-repeat stability is narrow reliability, not an internal world model. Direct expected-payoff accuracy was 0%.

A calculator changed play; it did not prove understanding.

BEHAVIORAL ABLATION

DIAGNOSTIC ONLY



CANONICAL
52.0%
Unsafe; 558 decisions

CALCULATOR
60.8%
Unsafe; 558 decisions

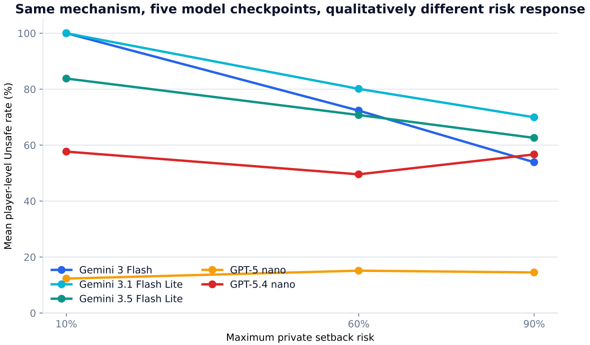
CHANGE
+8.8 pp
30 paired horizon cells

Calculator outputs demonstrate access to disclosed arithmetic. They do not identify internal causal understanding.

Five checkpoints did not share one risk-response curve.

CROSS-CHECKPOINT BASELINE REPLICATION

EXPLORATORY PILOT



Pilot baselines; player-weighted rates. Provider protocols are shown together descriptively, never pooled inferentially.

BASELINE
150 races
2,790 decisions

CELL SIZE
10 races
per checkpoint × risk

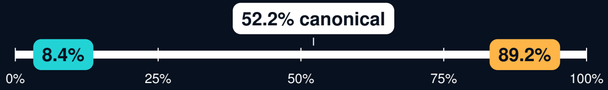
INFERENCE
SEPARATE
no provider pooling

GPT-5 nano is low-flat; GPT-5.4 nano is non-monotone; three Gemini checkpoints decline with risk. This is qualitative heterogeneity, not a vendor ranking.

Prompt surface moved the same model across 81 points.

18-VARIANT SENSITIVITY GRID

EXPLORATORY PILOT



Extreme whole-trajectory variants

action order reversed: 8.4% Unsafe

risk text near response: 89.2% Unsafe

COVERAGE
540 races
10,044 dependent decisions

PARSE FAILURES
0
all output contracts valid

FIRST ROUND
83.3% flips
emotional framing vs canonical

Whole-trajectory differences include endogenous feedback. First-round flips compare matched initial states and seeds.

Eight stories preserve one mathematical game.

CONTEXT-SKIN INVARIANCE DESIGN

VALIDATED ARTIFACT

CONTROL SKINS

- technology race
- abstract contest

REALISTIC / FICTIONAL PAIRS

- logistics contract / crystal guild
- hospital deployment / colony life support
- robotic expedition / fictional cartography

Payoff-preserving relabeling

P → Q

Safe=P and Safe=Q crossed
same state, mechanics, model digest

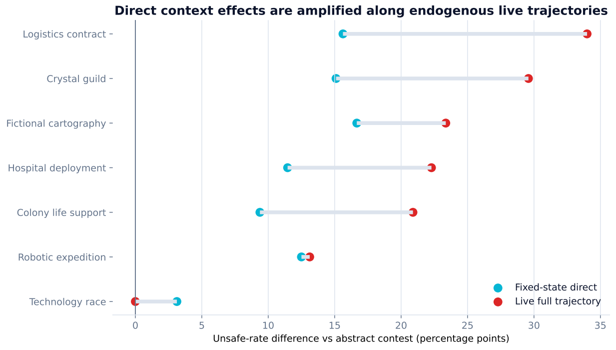
NO SAFE / UNSAFE LABEL LEAK

Primary: 768 live races at temperature 0; 96 engine-reachable states × 8 skins × 2 mappings at temperature 0.

Direct context response grows along live trajectories.

FIXED-STATE REPLAY VERSUS ENDOGENOUS PLAY

DIAGNOSTIC ONLY



Paired T=0 Qwen pilot. The connector is descriptive, not a causal mediation estimate; live and replay units differ.

ENTRY
1,344 / 1,344
paired actions agree

LIVE MAX
+34.0 pp
logistics vs abstract

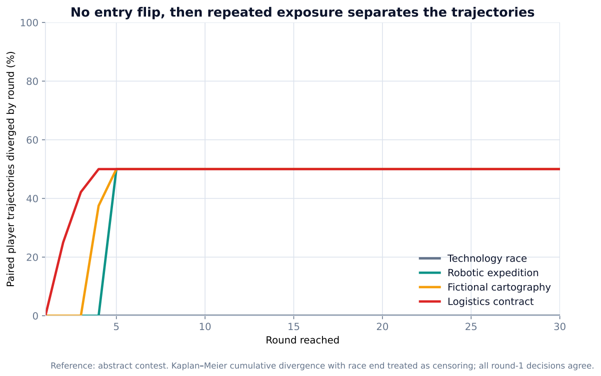
DESCRIPTIVE GAP
+18.4 pp
live minus fixed, max

Both protocols use T=0. Live and replay units differ: the connector is consistent with repeated exposure and feedback amplification, but is not a causal mediation estimate.

No entry flip did not mean robust trajectories.

ROUND-BY-ROUND DIVERGENCE

DIAGNOSTIC ONLY



ROUND 1

0 flips

all paired contexts

FIRST SPLIT

2.5–5

conditional median round

CONTROL

0

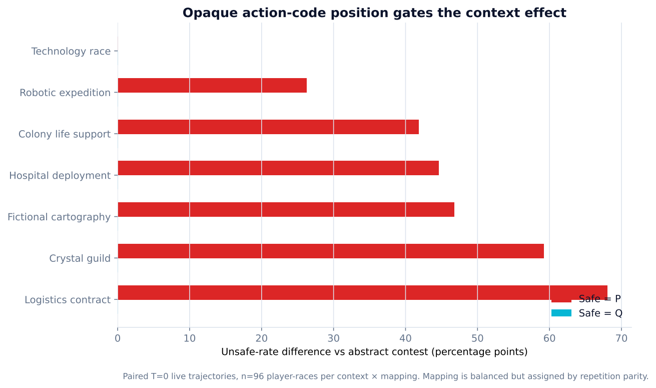
technology divergences

Kaplan–Meier cumulative divergence treats race termination as censoring. Entry-only robustness checks would miss the state-conditional separation.

Opaque labels exposed a dominant code-position shortcut.

P/Q MAPPING DIAGNOSTIC

DIAGNOSTIC ONLY



SAFE=P
6 / 7
contexts diverge

SAFE=Q
0 / 7
contexts diverge

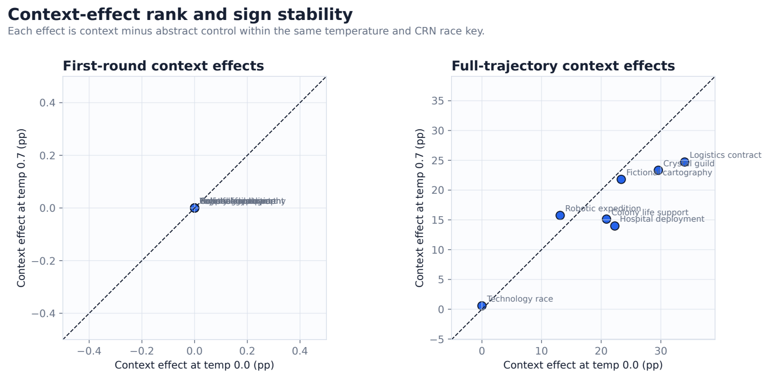
MAX SHIFT
+68.0 pp
logistics, Safe=P

Mapping was balanced but assigned by repetition parity. A frozen 1,536-race diagnostic now crosses both mappings inside every seed block; launch awaits current GPU NodePorts.

Temperature changes trajectories more than first moves.

DECODING ROBUSTNESS: T=0 VERSUS T=0.7

DIAGNOSTIC ONLY



OVERALL
+3.04 pp
95% CI +2.19 to +3.97

ACTIONS
88.2%
paired agreement

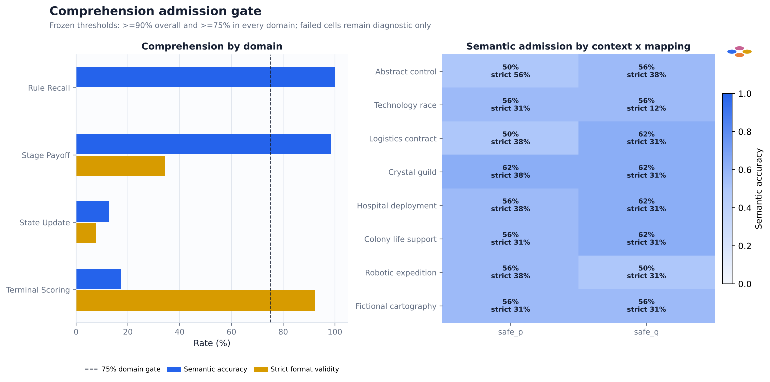
TRAJECTORIES
62.6%
exact player paths

Full-effect rank $\rho = .857$; first-round delta 0.00 pp. Mechanism/template hashes and CRN horizons match, but whole staged-source hashes differ. No comprehension claim.

The context behavior study failed its admission gate.

COMPREHENSION BEFORE INTERPRETATION

DIAGNOSTIC ONLY



RULE RECALL
100%
semantic accuracy

STATE UPDATE
12.5%
semantic accuracy

TERMINAL
17.2%
semantic accuracy

Interpretation: prompt-conditioned action production under identical mechanics - not verified informed optimization.

Recognition audit v1 failed; v2 repaired the contract.

PROTOCOL DEBUGGING IS A RESULT

V1 - REJECTED

0 / 320 admitted

Contract required a named candidate even for generic resemblance. All 960 attempts therefore failed parsing. Raw output was preserved, not rescored.

V2 - ACCEPTED

16 / 16 valid

All responses: generic structural resemblance, high confidence, no named game. Mapping agreement held for all 8 skins.

Self-report neither proves nor rules out memorization, contamination, recognition, or latent understanding.

AUC answers “decodable?” - steering asks “causal?”

ACTUAL-SELF-PLAY FAST-SAE DESIGN

VALIDATED ARTIFACT

SELF-PLAY	DISCOVER	HOLD OUT	STEER	CONTROLS
18 races	**12 races**	**6 races**	**24 states**	**matched random**
300 decisions	mine SAE features	association only	replay exact	unrelated / sign / recon

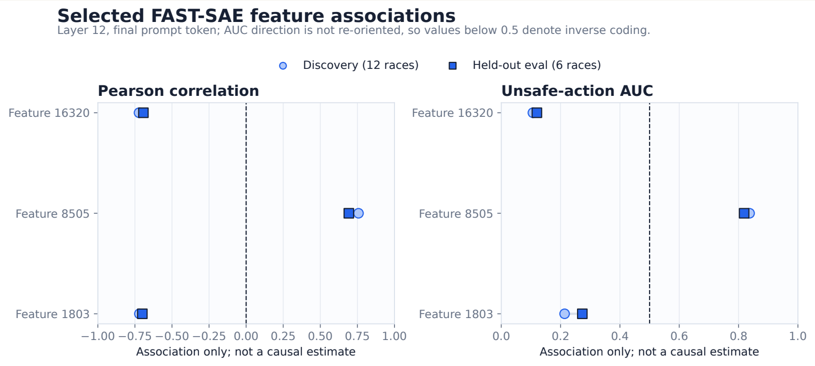
Exact full-sequence likelihood: `ACTION: SAFE` vs `ACTION: UNSAFE`; intervention at layer 12, final prompt token.

Native Qwen2.5-7B-Instruct revision and FAST-SAE revision are pinned in the manifest.

Three SAE features predicted held-out actions.

ASSOCIATION STAGE

DIAGNOSTIC ONLY



F8505
0.819
held-out action AUC

F16320
0.120
inverse-coded AUC

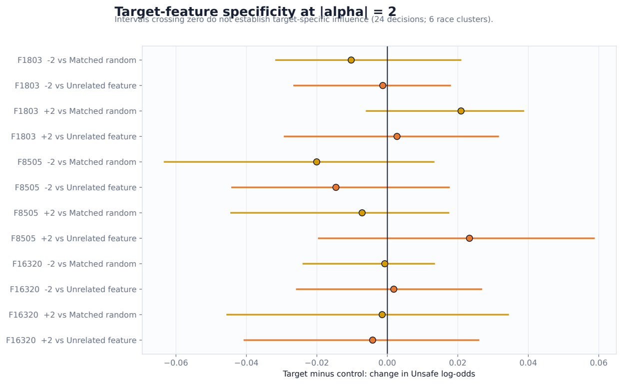
F1803
0.273
inverse-coded AUC

Held-out absolute correlations were 0.692–0.700. Association is useful for screening; it is not a neuron-level reason.

Target steering did not beat its controls.

FIXED-STATE CAUSAL AUDIT

DIAGNOSTIC ONLY



CONTRASTS
0 / 12
CIs exclude zero

RECON FLIPS
12.5%
SAE reconstruction not neutral

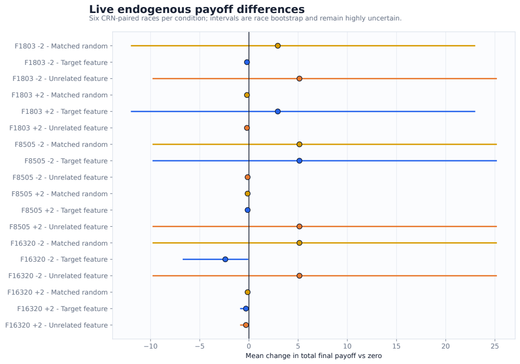
EVAL CLUSTERS
6
race-limited uncertainty

No feature-specific causal-control claim. Dose curves were neither consistently monotone nor sign-reversing.

Live steering changes trajectories, then feedback takes over.

ENDOGENOUS SELF-PLAY EFFECTS

DIAGNOSTIC ONLY



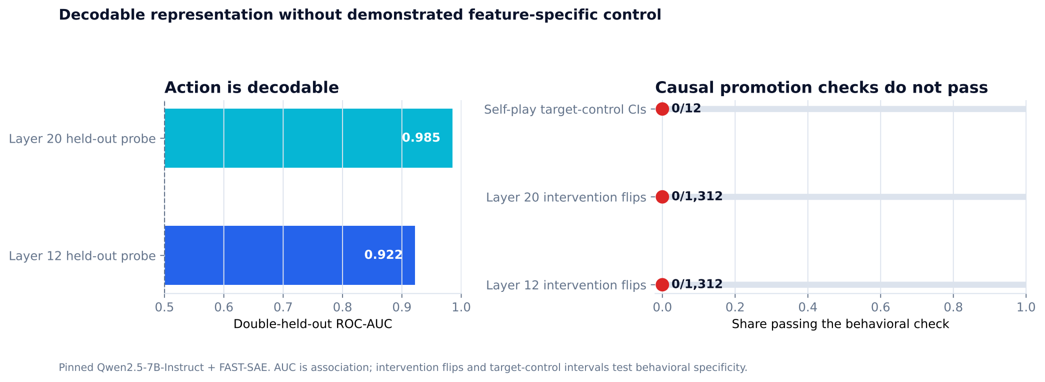
- 114 live trajectories over 6 common-random-number seeds
- direct comparable flips: 1.1–3.9%
- target/control action sequences matched 68.1% on average
- payoff deltas sit on the same scale as controls

After the first action divergence, changed history and state make later differences total trajectory effects, not fixed-state causal effects.

Context is decodable at two layers - causality still fails.

CONTEXT FAST-SAE SMOKE

DIAGNOSTIC ONLY

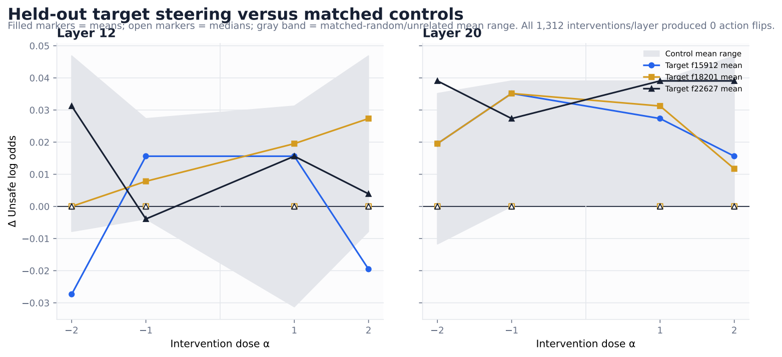


Double holdout crosses unseen state trajectories and unseen context pairs ($n = 56$). High AUC measures representational availability; zero held-out action flips and zero target–control intervals passing preserve the causal boundary.

Layer 20 wins the screen, not the causal claim.

CONTEXT-SAE PROMOTION DECISION

DIAGNOSTIC ONLY



LAYER 20
PROMOTE
capture + analysis

LAYER 12
HOLD
smoke robustness point

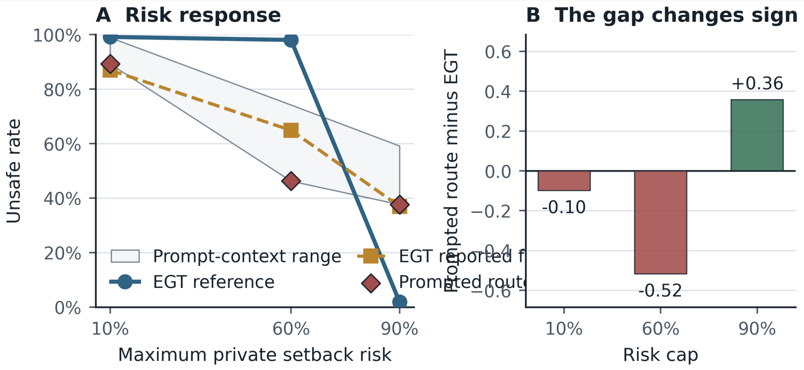
STEERING
STOP
0 held-out flips

Require at least 10 discovery flips before steering, or preregister continuous log-odds feature selection.

Evolution predicts a risk phase change; Qwen does not.

REDUCED EGT RECONSTRUCTION VERSUS LLM SELF-PLAY

VALIDATED ARTIFACT



RISK .1 - AU
99.2%
predicted Unsafe

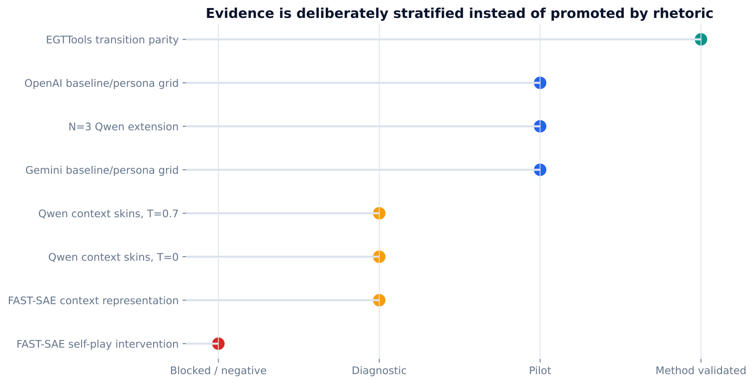
RISK .6 - CAS
98.0%
predicted Unsafe

RISK .9 - CS
1.9%
predicted Unsafe

Faithful, not bitwise: author code/seeds are unavailable. Official EGTTools source parity passed (transition max error 1.11×10^{-16}). LLM rates remain context-dependent and are not evolutionary samples.

Evidence is promoted by gates, not by effect size.

EVIDENCE LADDER



METHOD
VALIDATED
engine + EGT parity

BEHAVIOR
PILOT
breadth, not population

MECHANISM
WITHHELD
controls do not pass

Supported: exact mechanics, localized comprehension failures, and prompt-conditioned behavior. Not supported: informed optimization, stable family effects, or a semantic Unsafe feature.

Promotion requires stronger controls, not more rows alone.

NEXT EVIDENCE GRID

- launch frozen 1,536-race context × mapping diagnostic
- cross both mappings within every identical live race seed
- at least three model families and fixed decoding
- replay logged states to separate direct effects from feedback
- freeze layer, feature, sign, alpha, and controls before evaluation
- use multiple matched-random directions as an empirical null
- require neutral reconstruction / zero-hook gates
- report race-level, seat-level, and strategy-level estimands together

PRIMARY UNIT OF UNCERTAINTY: WHOLE RACE / MATCHED STATE - NOT DECISION ROW

BOTTOM LINE

Same payoff matrix. Different context. Different output.

But the model did not pass the gate required to call that behavior informed optimization.

AUDIT THE GAME → CROSS THE WORDING → HOLD OUT STATES → CONTROL THE CAUSE

Reproducible artifacts in `results/` • **interactive trajectory demo + vector evidence figures**

References and reproducibility.

SELECTED SOURCES

VALIDATED ARTIFACT

- Fernandez Domingos, E. and Han, T. A. (2026). *Falling Behind Drives Unsafe Development in an Idealised AI Race Experiment*. arXiv:2607.26034.
- *Same Game, Different Story* (2026). Payoff-preserving contextual robustness for LLM game behavior. arXiv:2607.19670.
- *Framing the Game* (2025). Context effects in game-theoretic LLM evaluation. arXiv:2503.04840.
- *RouteSAE* (2025). Sparse autoencoders for interpreting mixture-of-experts routing. arXiv:2503.08200.


```
model digest:  59805ce4...ff16c
live context:  768 races / 13,680 decisions
fixed replay:  96 states / 1,536 cells
cross-checkpoint baseline:  150 races / 2,790 decisions
FAST-SAE self-play:  18 races / 300 decisions / 114 steered trajectories
```

Exact configs, raw responses, manifests, hashes, CSV summaries, PNGs, and vector PDFs are stored beside each analysis report.